

ActInf GuestStream 122.1 ~ Radical Relation, Scaling without Hierarchies,

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Hello, welcome everyone. It is October 15th, 2025. We're in active guest stream 122.1 with Andrea Hiot, who will be presenting and discussing radical relations, scaling without hierarchies, models without meat. So if you're watching live, please write any comments or questions in the live chat. Otherwise, Andrea, thank you for joining. Looking forward to sharing and discussing. Thanks, Daniel. It's really good to be here. Sort of last minute. So this is very exploratory. And yeah, I'm Andrea Hiot, and I'm going to try and link some ideas kind of as we go. We'll see. We'll see how it goes. I just wrote a paper called Radical Embodied Relation at Any Scale, and it's coming out of the Radical Embodied tradition. I think you've had a few people on here from that tradition in the streams. But I'm not going to spend a lot of time looking deeply at the paper. But I am going to think about it in relation to active inference and also to this other paper, which came out very recently with some people in Heidelberg, from some people at the University at Heidelberg, which is where I'm at, and other places too in Germany. A lot of this comes from my time at Max Planck with the Dole Lab. And a lot of these themes are about the hippocampus and movement. And this paper, Grid Cells Accurately Track Movement During Path Integration-Based Navigation Despite Switching Reference Frames, is a really exciting paper. I know the title might not show that unless you know about all these things. But they're moving forward some of the things that I talk about in my paper in an interesting way. So if I can get the... Okay, did that shift? Yeah. And feel free to interrupt me anytime if there's any way to do to... You too, Daniel, because if I say something that doesn't make sense with active inference, I'd love to know. So I'm thinking about radical embodiment meets active inference. I actually think they're already really doing a lot of similar things. In the way that I present it in that paper. Of course, there's a million ways to think about all of this. But one sort of obstacle that isn't an obstacle for me, but that comes up a lot when I'm talking about Carl Furstner's work or about other work is this idea of the model being in the brain. And Carl and I had a talk about that on Love and Philosophy. And one of the things that most... A lot of people wrote emails to me about was that. Models in the brain. It's... And I know it frustrates Carl and others, but I think it's worth lingering on a little bit here because I think it's a kind of confusion that we can really get past. And maybe these two... There's a lot of different traditions, traditions actually. That I think once we move past that block, start to start making a lot of resonance. And that's one reason I talk about way making, which is another sort of annoying word I know for some people maybe because it sounds like a little bit sweet. But what I'm really trying to do with that is pull back from all these dualistic ways that we say we don't take part in anymore in terms of mental, physical, or mind, body, or whatever. Pick your dualism. Or body, brain, or mind, brain. But we still... Our language is still sort of

pushing us into those, which is often what's happening with this models in the brain thing.

So way making is just kind of trying to understand all of this action in this process in a way that we can assess from different points of view. And it's just trying to present an approach that gets us out of those routes of the binary that we don't always notice. And if you really care about that, you can find other ways, other places I've written about it. But this is just coming out of that a little bit.

And the way making is really just an approach. It's not some kind of rigorous thing. It's what I'm trying to do to break us out of those patterns. And then we can assess different forms of navigability in different landscapes. So they could be geographical, conceptual, emotional, etc. But I'm going to focus here on the hippocampus, which is what I studied when I was doing my sort of theoretical neuroscience degree, always connected to philosophy, and which I was really interested in about 10 years ago. And I started working on it. And a lot of papers were coming out of Max Planck, just beginning then to link what is now sort of being taken for granted. But this idea of memory and navigation as being connected in a deep way. I've had a lot of talks about that again on Love and Philosophy. If you really care about it, you can go there. But here, I just for the hippocampus, it's the seat of memory, you've probably heard of it as connected to memory. When you take the hippocampus out in these famous cases like HM, then you can't create new memories. Again, I'm sort of generalizing, but that's the basics.

And then you've probably heard of place cells and grid cells and this brain's GPS, which is the same part of the brain. But this is when we're moving through a landscape, how we're sort of finding our way. They're literally firing in relation to how we move. They're super cool. Like if you look at the actual experiments where you can see this real time activity in the brain and in relation to the environment, the place cells, grid cells, and so on. And then we have this idea of the cognitive map, which again has a million different views, but that connects back to that model in the head sort of thing, this map that's in their head, or we'll get to it, but the hippocampus as a cognitive map.

And none of these are really wrong or right, but they're very relative to that structure of that dualism I was talking about where we're making choices. So there's kind of three iterations. I just want to mention sort of briefly and in the hippocampus literature, and there's a lot in the hippocampus literature. So this is a very tiny part of it, but important part for this. The memory part, one of the most famous papers, 1957, you've probably heard of HM if you've studied like memory at all, the man who had his hippocampus removed from epilepsy. And there's this famous paper with Brendan Milner and Scoville coming out of this McGill school. And I'm going to generalize it a lot here, but from this work, which was a lot of very interesting, amazing work, especially kind of setting the way we think about representations and models in the brain, we kind of started to think the hippocampus has to do with memory because it's linked, obviously, in these studies. And it's more, you know, it's essential for the explicit memories. And, you know, once you've formed your memories, you can still have them, but you can't form new ones. And there's a lot of nuance about that, too. But that's just one iteration that's really important. So we started thinking of memory and now there's these kinds of memory, they're in the head, in different parts of the brain, and so on. And then in the 70s, John O'Keefe found these cells that I was just talking about, place cells, which fire in relation to a place in the environment.

And this is where this GPS of the brain idea started, which is a cartoon version to say it's a GPS. But again, something super cool is happening. And, you know, there's really something special going on in our body in relation to the environment and place cells, grid cells, border cells, head direction cells. And with the New York downstate guys, we're finding that around this time. So there's this whole beautiful, wonderful, exciting literature about the same area of the brain that we thought was essential for memory and still is. And for a while, people thought, okay, how do we reconcile this? It's memory and it's navigation. And like, how does this go together? And if you look back at a lot of the papers, they're trying to figure that out. And there's almost two kind of groups going. And in the late 70s, and over the past 30 years, this book, *The Hippocampus as a Cognitive Map*, has become sort of an orienting force in certain parts of neuroscience, which is just really spread and into all kinds of ways. I mean, from the computation to the cognitive neuroscience, which is thinking of Edvard Tolman's idea of the cognitive map in *The Hippocampus*. So then we have this cognitive map. So that's just the kind of iterations that we got now thinking about cognitive maps in brains and this thing I brought up about models and representations. And then what I was doing in 2000, whatever it was, 17, 18, 19, this was a time when a lot of labs like Tim Barron's lab and the Dolo lab at Max Planck, Tim Barron's is in London, were starting to rethink this connection between memory and navigation in a really interesting way. And again, there's tons of other people who are doing it in other papers, but they made some really important big papers for me, where they started linking this in different ways. And this idea that we think in the way we navigate, or we navigate in the way we think, or something like that. So for me, these papers, which I don't have time to go into completely here, but they started to show that we have this ongoing process, and we're assessing it from different ways. And you can't reconcile, you know, like when you're looking for something that we call memory, you don't reconcile that with what is happening when you're doing navigation. However, maybe we can sort of detach a bit in the way I was talking about with the way making. And this might offer us a way to reimagine what we're really talking about with these cognitive acts like memory and knowledge and the old way we thought of them. And these physical acts like moving through a room, finding our way through a city. So I just find this incredibly interesting and wonderful and the work that they were doing then. And my last paper comes out of that tradition. And it's philosophical, though, and it's talking about the hippocampus a lot, but linking it to ideas of radical embodied cognition. And not only that, but you know, mostly that and in the sense of that we don't need to think about representations or models in the brain in the way we once thought of them. And of course, radical embodied cognitive science, like Chimero and everyone, is often thought about as saying there's no such thing as a representation, but they're not actually really saying that. Or maybe some people are, but here's the way I see it. And the way I present it in that paper is that we can sort of have our cake and eat it too. There are representations, but they're not in the brain. And it has to do with this assumption I was talking about that we still talk about looking for them in the brain, even though most people don't actually think we're going to find something

called a representation that you can sort of, you know, find in the meat. And I hate that word meat, and I can't believe I put it in the title, but I did because, well, I'll tell you later why I did.

But then we have active inference, which I have no idea what time it is, by the way, but so you're going to have to just tell me. But I think a lot of the things that inspire me about active inference and Carl's work inspire me about all this other stuff I've been talking about. So I'm trying to find ways to talk about this. And just the basic ideas that seem to have entered the atmosphere from, I don't know if this was the first paper, 2009. I just sort of guessed, but maybe it was even earlier. But since then, I feel like we've been trying to look for this unified framework, you know, where perception, action, learning are thought of in a similar way to what I was just saying about memory and navigation in the sense that we're not separating them in the same way we were before. So sensing and acting and how organisms explore and exploit and epistemic and pragmatic, I see a lot of resonances in that in the way that I'm trying to talk about memory and navigation. But again, this is just exploratory. But I do think it relates a lot to this confusion about models in the brain and the way a lot of people just turn off from either of these ideas, radical embodied cognition, or maybe even active inference when they when they hear that, or when they look at the way that's expressed in certain ways. But I actually think this is really exciting, that these are doing something similar. And I feel this excitement, but I don't know how to express it really yet. Because I just started thinking about it. And I'll try to I'll try to I'm going to unpack this more in another talk later. But this is just from today, really. And the past few days, thinking about this a bit. These obstacles that get in the way in this language and, and what it means, and I do think it actually means means a lot. And an interesting side note, with Lynn Nadell, who's one of the people who wrote Hippocampus as a Cognitive Map. In one of our talks, we've had like five now, I think most of them are public, he actually starts talking about active inference or about Carl's work and saying that when they were writing the book, they were thinking about a lot of similar things, they didn't use the word inference, and they didn't think of it as prediction. But, you know, he brought that up to that there was some kind of resonance there that that he felt. And, yeah. So if we get to this radical idea, this idea that we don't need representations, or that there aren't representations in the brain, for example, which again is one part of this tradition, which says a lot more than that. But I think it also relates to ecological psychology. I mean, obviously, that's a big part of that tradition, the radical embodied cognitive science tradition. And I feel that there's a lot of resonances here with all that I've brought up, that are helping us understand a different way of thinking about cognition that relates to a lot of work that's going on in biology, such as, you know, Mike Levin and Richard Watson and this cognition first and multi-scale. And I feel like tied up within that is getting out of these habits that we don't notice we have about dualisms and binary ways of separating what we're working on as we're doing the work, and as we're doing the papers. And one big glaring kind of thing, aspect of that for me, is the way we use neural representation, also mental representation, but I'm focusing on neural representations mostly in the paper. Because most neuroscientists, as I said, aren't looking for little diagrams or hieroglyphs,

as Romain Bressa calls it, in the brain. However, we write our papers as if we are. And I think that's causing more trouble than we... I mean, a lot of people are trying to figure this problem out. So obviously, everyone knows this is causing trouble. But so in that paper, I talk about them as communicative, I basically say we have representations, obviously, but they're not to be found. And that we have to understand that this is where it gets tricky for me, there's a difference between the way we're assessing a certain situation, and what we're using to do that, and what's actually in whatever it is that we're looking at as we assess. So that's what the paper goes into in a lot of kind of detail. But the reason I use this meat machines thing is because Ned Block, who was kind enough to let me sit in on his lectures, was it last year? I don't know, came out with this paper, Can Only Meat Machines Be Conscious

recently. And I actually think all these same subjects have a lot to do with what he's saying in that paper. And it's a good way to think about it, like that we don't actually have these models and representations in the body in that way, in that sense, but they're still really there. And what I like a lot about the grid cells during path integration paper is that it even goes another step further. It's a really recent paper, but it actually says like, okay, it's not a GPS in the brain, it's doing this, and I'm sorry for those who wrote the paper that I can't put this in perfect words yet, but it's, there's a lot of things going on at once with this system in the hippocampus. It's more constellatory, or like a constellation in my way of saying it, many dimensions at once, not just a GPS at the same time. So there's a way to have the models and assessments that are real, and they can be right or wrong, and they can lead to real results, but it depends on the task and the way you've set everything up. And I think we all know this, but really making it a foundational part of how we talk about it is kind of what we're trying to move towards, I guess. Daniel, what time is it? Do you? 22. 22. Okay. So I'm just going to go through some slides, which I just helped, Claude Coding helped me make these. So I'm just want to say that I just basically was saying what I wanted to try to show with them. And so we, if we think about the hierarchy of, you know, this is higher and lower and the higher is the reasoning, the planning, the remembering, and then that's all that stuff of memory and knowledge, I guess. And then the lower is usually thought of as the movement, the sensation, the action, the finding your way through the city or the room. And I think those don't really help us anymore in the hierarchy way of thinking about it. And I know I'm not new saying that. I think all of these traditions I've just expressed are saying that. But it's just a good way to look at how these are connected. So with the place cells, they're not representing. We were thinking of earlier, it was almost like there's a cognitive map in your head that's, you know, guiding you or something. But all of this is actually happening much more in like an ecological sense or an immediate sense or an active inference sense, actually. And this, you know, where it's where it's ongoing, it's real, it's happening, but it's changing all the time and in many dimensions. So it's not that there's one little kind of hieroglyph that's always happening when you encounter something, but there's patterns and regularities that can be assessed. So, you know, there's, these are just different ways of thinking about that. They're more like predictive dynamics that are bound with

the movement, we could say. I'm just thinking about ways words might go across all these different traditions that I've just brought up. And then the grid cells, like in the paper, I didn't go into it, but the grid cells are kind of the biggest mystery and the most wondrous thing in the Moser lab. They're doing all kinds of really cool stuff with it too, which I would like to introduce with this paper, because I think there's resonance, but that would be a whole other talk. But they have these hexagonal firing patterns and it's not like a single coordinate system. At least as I've read this paper, it's starting to think about it as like doing multiple reference frames at once. So it's anchoring to different tasks. And in that way, I expressed with the GPS, this immediate sort of way that you can see that the body is, you know, synchronizing with its environment to put it in too general of a term, and you can actually see that and, and you can predict it, you know, you can watch the rat in the maze, and then you can predict exactly what is going to fire after you've done that. So there's this definite regularity and kind of amazing thing that's happening. But it's happening at multiple levels, and it's not happening in a way that clearly can be understood as here's the mental and here's the physical, or here's the moving through the room, and here's the moving through the memory. It's more about which one are we looking at, which one are we assessing, and what are we assessing about the body relative to whatever we've, you know, assumed about that word, whether it's mental or physical or whatever. So just one way of thinking about it, I was thinking about like, where are models? Where is the model? And I'm being really sloppy here with the word representation and model. I know that, but I think the idea is similar of what I'm trying to say. So when you ride a bicycle, where is the knowledge? Where's the, you know, of course, that's a different kind of memory. So I'm again, mixing all of this, but you might say it's like stored somewhere in your body or in your cortex. And I could have had a better example of this, but just the idea that you're doing something and you've stored it and you're bringing it back out. And, but what actually is really happening, obviously, is that you're coupling with the bike and with the environment, and that whole process is the memory. So yeah, you have a memory of what it's like to be on the bike and move it, but that's changing as you move through the landscape depending where you are, depending what your body is doing that day. And you might have the same bodily patterns or the same neural patterns, but it's the riding competence. It's, it's the communication of your body, the bike and the environment that is the pattern that we're going to notice that continues. And there's a lot of different ideas about models in the brain that I thought I could go into and I'm just going to skip that. But basically this model emerges from dynamics of coupling, not from the neural structures. So coupling, I don't know that word, I'm trying to think about C's, but it's more communicative communication. I don't know if that's the right word, but because coupling again, it's dyadic and it's dualistic and it's two. And when you're on a bicycle, you're constellatory, you know, you're coupling with lots of things at once. And all of those things are going into the system that we're calling that whatever is happening in the hippocampus, but also in all these other ways. I think active inference obviously already shows this in a way there's not prediction errors living somewhere there, you know, in the dynamics of the levels. But yeah, I mean, this is where it gets really tricky and a lot of people could get very mad at me when

I'm thinking about these ideas in a big way and then the actual maths of them or the language even that we're using right now to describe them and how those get very confusing. So it's not, and I'm sorry, but I don't think that it's that the math is in our brain or that the math is in the meat, you know, or that the language is in the brain or in the meat or that any of this needs to be located in that. But it's more of that's how that's one more level of exactly this process, which is why it's so confusing. So when we're using language or math, we've, we've, we've sort of externalized and created a representation externally so that we can share that and couple together via this. So as to better understand one another's experience, I go into that more in the paper in a more precise way. But so we still write and talk about models and representations as being in brains and in heads, even though we know they're not. And we're separating what's not separable so as to understand it. But then we forget that it wasn't separable. And we've talked about it as if it was inside. So this idea of holding paradox is somehow, I mean, it hurts a bit to try to think like this, but it's not that I have a thought and it's in my head, and then I encode it in language, and then I describe it to you in words, and then it's in your head. And then, you know, you have it, and it's your representation.

It's not, there's not a thought that I'm giving that any of us are giving to each other in the same way that in the GPS system, or the whatever we want to talk about what's happening in the hippocampus, there's not some little hieroglyph that I could have and just give to you. So it's more this radical way of thinking about this, which I think community, it's communicative, even if it's communicative between you, what you think of as yourself and your body or parts of your body.

Still, language sort of forces us to go back to talking about models and representations and brains.

It's really hard to talk about this. And if you look at all the neuroscience papers, we still talk like this, even though people, a lot of this is common sense, as I discuss in the paper. But why does it matter? I think it matters a lot. I didn't go on my soapbox, I was going to earlier, but because we're all sort of talking past each other in ways, and I feel like we're getting a little bit caught up on especially, at least theoretically, on things that are exciting to argue about, but aren't actually helping us understand how to communicate better, how to understand what's hurting us and others, how to care about that in a deeper way. And I really think so many people are doing so much amazing work in all these different fields. And then we end up trying to prove to each other that our theories are right, when we're kind of saying the same thing, just with different language. So I know that's a little emotional, but I do think all of this actually really matters because a lot of the ways we're just coming at things are, are blocked. We're not seeing the resonance due to this. So this would be probably too much to go into. Do I have like five minutes or something now?

Yeah.

Yeah.

I guess if we understood models as communicative, or I was trying to find some kind of active inference language, but then it's not that the math has to change or anything really has to change. What changes is just our way of talking about it. And that is hard. A lot of people who I've talked to this about say that they know these things, like it's not that they're looking for models and brains,

but that they can't help but talk about it in that way. So I wonder how we can sort of shift that to thinking of this communicative, relational model in a way that would be easier to discuss, you know, in our language. Yeah, I don't know. So I was thinking about language, of course, because that's a very fascinating subject. But I was thinking about like language coordinates as mutual prediction, or we're not transferring information, but we're sort of scaffolding each other's paths of communication in that conceptual space, when we're having a conversation and language is a set of regularities that we're sharing in that third way that I sort of expressed that helps us do that. So it's like sharing. It's, it's, it's another kind of landscape that we've built that we're sharing, and that is conceptual. So if we can start thinking in these dimensions, which is, you know, kind of fractal and wild, instead of just it's mental or it's physical, we can start realizing, gosh, there's so many landscapes, we have to realize which ones we're discussing, what regularities we're discussing, and how we're sort of scaffolding one another's way of entering those landscapes. Because often, I feel like we're on the edge of some sort of breakthrough in a lot of our work, but we're not letting the data resonate, because we're not realizing that we're scaffolding each other out of each other's landscape somehow. Yeah, so how could we sort of coordinate those trajectories and think about it that way? It's just kind of a wild way to think about it. Last thing is like giving directions. So when you give directions, you're, you're not transmitting a representation, you know, I'm not giving you like my cognitive map, but you are sort of co scaffolding, you're helping someone else scaffold what you've already experienced. So in a way, you are sharing a way, you're helping scaffold their irregularities. Yeah, and I think I will leave these, these ideas for another time. But I was thinking about reshaping attractor landscapes as, you know, in language, there would be a lot to go in into that. But it's just exploration, thinking of it in this way, trying to get a little bit almost psychedelic in the sense of getting out of traditional ways of thinking about space, and get in so doing, realizing we already have a whole lot of similarities across a lot of different traditions when it comes to trying to understand cognition, and that that's actually very connected to the ways we move through space already. And this might be one way that we can start to think about some of this. Yeah, that's just thinking about what we mean by space and landscape differently. And I had a baby example, but and then this constellatory thing, I don't know if it makes sense to anyone so far, people aren't really sure what the heck I'm talking about. But I'm kind of envisioning instead of this either or dyad, that we try to understand things more constellatory. And of course, it depends on where you're going to center that. But you know, there's constellations and constellations and constellations. And a lot of systems theory, and of course, graph theory and everything else, is doing a lot of that. But I still try to find a way to envision that and think about it. And maybe language or communication, like understanding that these are different ways of communicating in different spaces between different parts of us, our bodies, or different bodies, feels like to me, something to maybe explore this constellation of organism, environment, tools, culture, etc, etc. And I was thinking of jazz as a kind of a great example of that, as always, you know, typical example,

but just to bring in some music for the end, that this is a, you know, we all bring in our instruments, our language, and we create something in real time coordination that is, for me, a kind of constellation, a sensory constellation of all the people and the ways they're experiencing it and of that room in that moment. So yeah. Constellatory, I guess, is the way to leave it for now. Yeah. Okay, I'll just leave it there for now. I think that's probably the end of my time. And I have to go around eight. But maybe that made no sense at all. But at least I tried to put some ideas together. So thanks. You put many stars in the constellation.

That's all I need to do. That's all I wanted to do.

There's a lot to reflect in there.

Indeed.

I was struck with the laterality of wayfinding and how that sort of branch swinging or movement in the cognitive

or material spaces, it has a sort of flatness, which alludes to the without hierarchies from your title and actuality. Like whatever association a given cognitive entity makes in a moment, no matter how comical or useful or absurd or taboo in a given situation, the association is always actual. Whereas when things are thought of in terms of the induction and deduction, where the jump is from some phenomena to either a component or to a category or a type of thing, then it gives a different grip on it.

And so there's sort of a...

And then you turn that not over in your work and in this presentation from many sides with what language can and cannot do and what it enforces, what it enables.

And that goes from the whole worldview scale, the kind of paradigmatic within professional science, and also within what's more broadly received, all the way down to the very syntactic, like between is dyadic, but among is three or more.

And in is an enclosure preposition, but with is an instrumental.

So it really does make the last mile in our utterances and in how our inner and outer landscapes are conceptualized of and navigated, which is everything.

Yeah.

And it has that historical...

I mean, when you were talking, I was thinking of the links of it, everything we've been exposed to in those terms you just expressed before the moment will be part of that moment and differently for each person or each, you know, each whatever, whatever group or person might be in question.

I guess the question that I wonder if you have any insight about is...

how do we start to think about the space of language as something where...

I don't know, it's almost like a meta communicative that we're...

we don't...

I mean, even when you were just describing the dyadic or the way that we use language and so on,

how do we think about that as...

as its own landscape, even as we're in that landscape? Does that make any sense at all?

Is it too much to kind of...

I don't know, is it even helpful to be aware of it like that on a mathematical level or on a scientific level?

Or...

Because, you know, there's been this debate in neuroscience about like we use words from William James that we just sort of look for in the brain.

Buzaki brought it up and stuff and a lot of people have debated that of just do we need new words and so on?

And I wonder if anyone in the active inference world that you've talked to or that comes to mind has had any insight about...

Because these are kind of new ways of talking that are coming into the... into... into the science.

And yeah, I wonder if that's helpful or that's...

If you've, you know, if you have anything that comes to mind when I just even bring that up, actually.

Yes, a few...

A few pieces.

One is the know that and know how and other sorts of distinctions that have been made in cognitive and linguistic philosophy.

Like the more phenomenological side of it or are you...?

All like propositional knowledge and different ways of knowing.

So...

Mm-hmm.

Giving a structural framework to help get handles or...

or context on what kinds of things can be expressed through language, what kinds of things are embodied skillful performance.

And...

Um...

Even...

Uh...

Earlier today, Robert Warden presenting on a sort of architectural approach which Elliot Murphy, other...

other people have sort of laid out pieces where there's a primary ingress layer from the phonemes in heard speech or edge detection visual aspects of reading.

And kind of a primary ingress into a more conceptual Bayesian graph structured hierarchical generative model.

Like when you said that the language models and the world models are not different.

And then...

Rather than a sort of unopinionated large generic architecture,

transformer-based or neural network-based, where a given input has a given embedding or state-space representation.

And then that state-space representation can be used to go like from words to words or from words to images, things like that.

All the kinds of AI that we see.

Rather than having that hyper-parameterized, unopinionated architecture, what biological systems with their sparsity conditions seem to do and what points towards alternative, more cybernetically informed architectures would be the structure of that state-space is somewhere between that generic and a very, very opinionated, very sparse, much smaller state-space where samples could be pulled or representations could exist in that state-space in a moment, which could be used to propagate back out towards language through introspection or self-report.

Or that state-space representation could be used...

So that would be language emissions or action planning.

So it's like if I have situational awareness of the cup to my right,

I could perhaps implicitly or explicitly explore what would happen if I moved it here and verbally describe that happening, both coming from a common representation.

And then I think what you are importantly bringing the empirical papers and work to the fore is, okay, GPS is a metaphor.

We could go down that rabbit hole with what metaphors are, but it isn't a static, allocentric overlay.

If we're going to get into that first-person

way-making and way-finding, then there's simultaneously like an egocentric reference frame, and then a sort of counterfactual egocentric.

What would I see if I were there?

And how that plus heuristics, habits, ways

can address problems that otherwise would be approached for more of a brute force, calculative way.

But like seeking higher ground and looking around could be a heuristic on a landscape that doesn't go through the policy rollout of every single location to go to, and what would I expect to see at every location? That might be solving a problem in an exhaustive way where the first-person perspective and heuristics give an alternative approach.

But that first-person heuristic, it's not doing those rollouts, so it doesn't have access.

So then that's part of this kind of disciplinary approach crossfire where then we're seeking those policy rollouts in neural data, but they aren't necessarily even there because that's reifying what the models, what the program's architectural opinionation is.

Yeah, what you're looking for too, or what manifold comes.

But curled up in all that you said, to me I feel like we confuse...

Like the LLM and the AI side of all this is so important, right?

But in a way we don't understand that those are not...

That's a different umwelt, is what I guess I'm saying.

Like if we could understand these as different layers, we have this language sort of umwelt, or this conceptual space that we now share in a really profound way with machines.

But that's one layer, you know? And instead of like saying, okay, that's mental, and so everything that we've ever talked about is mental now is attached to this technology.

It's just like I'm trying to kind of, you know, manifold this out or separate, layer this out so that we understand we're studying, we're sharing a conceptual dimension now.

A language-based dimension. That's really profound. But it's one dimension um that already maybe depends on certain affordances that we're bringing to it. So when you're talking about that phenomenological, oristic, or that that moment, that's why I really love trying to go back to the movement itself. And of course we could talk about robotics and all that we've learned from movement and robotics about, you know, technology and AI. But if we just think about, if we stop thinking of it as trying to fit it into categories, and we think of it as a continuous process without beginnings or ends, and by that I mean whatever we're studying as cognition, then we start to understand how whatever position, from the moment it is a position, when you're assessing it, as it's moving through whatever regularities you're defining as the landscape, is building something like these patterns, which we can study in all these different ways, and which are constellatory or I don't know, like kaleidoscopic, or they're sort of layered in these fractal ways that are really hard for us to get a grasp on. So like we definitely are sharing some umwelt with technology in a new way right now, based on language. But I feel like so much of that is just getting, like you know, we're just taking on all this other stuff that isn't actually part of that umwelt, you know, in terms of everything we think of as what mind and cognition is from a human point of view. Does that make any sense at all? It's part of why I'm trying to find a way to separate out those things a bit.

What kinds of experiments or inquiries, if you had an infinite lab or kind of a dream setup, what would you want to see? What kinds of measurements or experiences or setups do you think would help move beyond checking for new neuroscience papers and unifying them within cognitive frameworks? I think if we could get away from the stimulus response kind of set up in some way, there's some, you know, I mean that for me is already based in this dualistic dichotomous language that's in a way that's very hard to get away from where we, the way we approach it already, where there's a subject and there's someone looking, maybe one way to do it is to kind of some find some way to think about the space that we're in as the third part of that and that we have to somehow in our experiments metricize the space, the regularities, the landscape, whether it's the room or whether it's language if we're doing it conceptually, if it's a conceptual experiment or something that I think there needs to be instead of the stimulus response, there needs to be some, at least start with a third, the third as a way into kind of this constellatory and whether that

means metricizing or thinking about the space itself, which is already kind of getting into some ecological experiments, I would have a lot of ideas about how to start to do that. The problem is my math skills and my, I mean, I can code and stuff, but like really coming up with different ways of visualizing this, I mean, I've kind of desperately tried to reach out to, because I think the first step is more, how do we visualize it? All of our visuals right now are still very hard to like express that there could be so many different shared dimensions. How would we, how would we, I mean, I'm thinking of graph theory or something, or these other ways that, you know, even just small worlds theory and the way we've tried to visualize things like that, but it would have to be sort of multi-dimensional where you could literally sort of somehow feel as if you can visually express how you, Daniel, for example, are part of many different landscapes all the time, and your body's reacting to that in one continuous way, but when you're going to study it, you would break it down into different regularities so, so you can understand it. So what would be the, the world that you're sharing linguistically? What are those regularities when you're talking with your family, when you're talking with a chat bot or whatever? What is that space? How would you mestersize that space in a way that could give you

some data so you could see what the differences are? How would we even, I mean, this is a really crazy idea, but you said unlimited. How would we sort of blockchain? I mean, how would we understand, like if we took, we already do it in a way with Google and looking at words and how often they're searched or something, but how would we like create these kind of trajectories for, you know, patterns of language and how, you know, because right now we use chat bots and they're basically pulling on everything that's ever been said or written by everyone, you know, that's online, but how could we follow those trajectories of where that's coming from and how that iterated into this, you know, things like that. I just, I feel like it, it's possible now with technology, but way beyond my skill set for how to kind of do that. But somehow combining that way of understanding the different landscapes we're in and their trajectories into, you know, how they've evolved to that point and understanding how we, how we are in them multiple ways at once, which again is all just a model. It's not the actual ongoing reality because that's not, you know, we don't, I'm not saying we could ever, I don't think we could ever have a twin of that necessarily in the models, but maybe.

I, I hear you on that. It feels like so many of the pieces are coming together.

With agentic approaches to representation and the way that those kinds of small scale operations in their toy

form all the way up on through extremely impressive are, are being increasingly described, which gives like an operational grammar for that process-based ontology.

So programming as, as, as orchestration and even implicit within object orientation are transformations and changes to objects. So again, recapitulating a sort of noun and verb non superiority. They play complementary parts in, in system expression.

And dealing with those first principles, with those radical aspects of situationality, like whether it's 4E or just perspectives, perspectives, all those kinds of features that

seem to fall out at the same time of that knot of engagement and the dawning of, of systems that let us describe and then see that happen in warp speed rather than just imagining it play out or, or, or doing it manually. Definitely.

You, you, you, you started to do it a bit with the ants paper, the last, I can't remember the name of it, but there's some resonance there, I think, in what you were, what you were doing, but I don't remember the name of it now, but.

In the inference paper, the, one of the pieces that we explored there was just collective behavior with stigmagy, with the niche modification. Otherwise there's the, the landscape of collective behavior is entirely transient, like the, and it doesn't leave a trace, but then if you have multi-scale activity within a space that is also changing, like people are emitting words into text corpuses that change the frequencies, then it's more like the, the balls are rolling on a trampoline and they have their own directions, they're changing the, the space as they move through it.

Well, what, what should we be on the lookout for, or just, what would be exciting in this last bit of this year? How do we, how do we bring, how do we bring something timely and contemporary to, to a lot of these questions?

Yeah, I guess, bring it into our everyday life, wherever we are, you know, how can we, whatever we're working on,

try to notice if we're thinking of it in, um, traditionally either or terms, um, which we probably are, because I think most of us are sort of set up to do that and most of our systems are set up on that, and maybe letting ourselves dream a little bit of what might be possible if we introduced a third or a fourth, uh, way of, you know, perceiving that. I mean, I think Johari windows and the fourth and a lot of different, um, traditions and the third, of course, there's so many ways we've been trying to do this, but I really feel like a lot of it's going to be practiced in kind of everyday life, uh, in noticing that we are still seeing things in a very particular orientation and that that's not wrong, but maybe we could open up our sensuality, uh, even as scientists or as philosophers or whatever we are, even if, you know, we were, whatever we're doing, uh, I find that like a really rich way of staying grounded in these times and of remembering there's a lot of possibilities and that we can open, like new worlds can open that we didn't expect, uh, just through perceptual shifts.

I don't know if ants have that happen to them, but I would imagine that they do.

Where that takes me is remember and dream.

Remember and dream. Yes, at once.

Sometimes it's the same thing. I mean, memory and imagination. That's part of, I didn't get into it, but that's actually part of what you start to see. They're very, very similar.

So yeah, remember and dream. That's very good. Let's, let's, let's leave it at that.

Thank you for always bringing to our space and institute so much scientific and poetic and fun, fun videos and writings to always enjoy.

Thank you too, Daniel. I love your work. You know that. I really appreciate you. Thanks for

letting me go a little wild on this one. It was fun.

See you. Thank you.

See you. Bye.

Bye. That was fun. Thanks.

Bye.

It's rough right now. Yeah.